

Jesuino Vieira Filho

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Education

MSc in Computer Science (AI Specialization)

Université de Montréal

Sep. 2025 – Present

Montréal, Canada

- o Average grade: A+ (ongoing)

BSc in Mechatronics Engineering

Federal University of Santa Catarina

Aug. 2016 – Sep. 2022

Joinville, Brazil

- o Average grade: 8.68/10 (2nd out of 40)
- o Completed two graduate-level courses in data science and machine learning (Grade: 9.5/10)
- o Bachelor's thesis: "Comparison of machine learning methods for short-term urban water demand forecasting in a coastal tourist city" (Advisor: Pablo Andretta Jaskowiak; Grade: 10/10)

Research Interests

representation learning, world models, generative models, geometric deep learning, ai4science

Professional Experience

Data Scientist & Software Developer

IconPro GmbH

Aug. 2021 – Apr. 2025

Remote

- o Led the integration of AI algorithms into *ares*, a software for optimizing manufacturing process
- o Applied research on reinforcement learning for optimization and improved distributed data pipelines

Data Scientist & Software Developer (Intern)

IconPro GmbH

Feb. 2020 – Aug. 2021

Aachen, Germany

- o Supported the development of *apollo*, a predictive maintenance software for industrial machinery
- o Reviewed and implemented time-series forecasting methods from research literature

Research & Teaching Experience

Undergraduate Research at LISHA

Federal University of Santa Catarina

Aug. 2018 – Jul. 2019

Joinville, Brazil

- o Evaluate deep learning models for people detection in indoor environments using OpenCV

Teaching Assistant for Programming II

Federal University of Santa Catarina

Aug. 2017 – Dec. 2017

Joinville, Brazil

- o Course focused on data structures and sorting algorithms taught in C
- o Assisted ~50 students through exercises and office hours, reinforcing topics covered during class

Publications († Peer-reviewed, * Equal/first author)

† *Toro-Hernández, F. D., ***Vieira Filho, J.**, & *Cabral-Carvalho, R. M. (2026, in press). Characterizing human semantic navigation in concept production as trajectories in embedding space. *International Conference on Learning Representations*.

† ***Vieira Filho, J.**, Scortegagna, A., de Sousa Dias Vieira, A. P., & Jaskowiak, P. A. (2024). Machine learning for water demand forecasting: Case study in a Brazilian coastal city. *Water Practice & Technology*, 19(5), 1586–1602. <https://doi.org/10.2166/wpt.2024.096>

***Vieira Filho, J.**, & Jaskowiak, P. A. (2020). Comparação de métodos de deep learning pré-treinados da biblioteca OpenCV para detecção de pessoas em ambientes internos. *Revista Eletrônica de Iniciação Científica em Computação*, 18(4). <https://doi.org/10.5753/reic.2020.1766>

Further Education

NeuroAI <i>Neuromatch Academy</i>	14 Jul. 2025 – 25 Jul. 2025 Remote
Spark on Kubernetes <i>Data Engineering Academy</i>	17 Oct. 2023 – 18 Oct. 2023 Remote
School on Complex Networks & ML in Neuroscience <i>ICTP South American Institute for Fundamental Research</i>	22 May 2023 – 26 May 2023 São Paulo, Brazil
Writing & Publishing Scientific Papers <i>Federal University of Santa Catarina</i>	12 Jul. 2021 – 20 Aug. 2021 Joinville, Brazil

Honors & Awards

Bourse d'exemption UdeM <i>Université de Montréal</i>	Sep. 2025
Outstanding Academic Performance <i>Federal University of Santa Catarina</i>	1st Semester, 2020
Scientific Initiation Scholarship (PIBIC) <i>National Council for Scientific and Technological Development (CNPq)</i>	Aug. 2018 – Jul. 2019

Selected Skills

Machine Learning Libraries

PyTorch, scikit-learn, Apache Spark, OpenCV, pandas, NumPy

Infrastructure & Development Tools

Docker, Kubernetes, Git, Linux

Programming Languages

Python, SQL, C++, C, LaTeX

Languages

Portuguese (Native), English (C1), German (A2), French (A1–A2, in progress)